Why spectrum-matched stochastic closures fail the elliptic test: operator-level pressure–temperature decoupling across scales

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Abstract

Weather and climate models rely on stochastic parameterizations to represent subgrid processes. A standard assumption in these schemes is that matching the energy spectrum—the variance on each wavenumber—captures the essential effect of unresolved dynamics. We show this assumption fails for the pressure field. The fast (elliptic) pressure correction in incompressible flows is a global, boundary-aware constraint; a spectrum-matched, phase-randomized surrogate with the identical power spectrum leaves the divergence constraint unrepaired by four orders of magnitude (RMS divergence 3.7×10^{-2} versus machine zero 2×10^{-15}). The surrogate test thus demonstrates that spectral equivalence is not dynamical equivalence for elliptic constraints: matching the power spectrum of an elliptic correction does not reproduce its action as a constraint. We establish the operator split driving this failure across five independent settings: field data show temperature tracking local diurnal heating while pressure carries a planetary-scale semidiurnal tide; direct numerical simulation shows pressure and buoyancy share the integral scale but differ by a factor of 2.4–4.5 in Taylor microscale, with the gap widening with Rayleigh number; a controlled linear-acoustics solver shows the hyperbolic-to-elliptic crossover scaling as $t_{90} \approx 2.3L/c$; a nonlinear compressible solver shows the fast-clock energy fraction scaling as M^2; and NCEP/NCAR reanalysis shows the divergent (fast) wind carries $\leq 4\%$ of kinetic energy, with a minimum at the classical level of non-divergence. These measurements establish that the operator split is a robust, measurable property of real flows. The surrogate test then proves that spectrum-matching alone cannot reconstruct the elliptic geometry—a hard constraint that stochastic closures must respect. Scope is explicit: the demonstrations are two-dimensional and periodic; no 3-D regularity claim and no forecast-skill claim is made; reanalysis is a model-assimilated confirmation at $l \leq 35$.

1 Introduction

Weather and climate models at finite resolution require closures to represent the effect of unresolved subgrid dynamics on resolved scales. Among the most widely used approaches are stochastic parameterizations that augment deterministic tendencies with random forcing, typically designed to match the energy spectrum or variance of the unresolved scales (Hasselmann 1976; Berner et al. 2017). The implicit assumption in these schemes is that energy matched on each wavenumber captures the essential dynamical effect of the unresolved processes.

We show this assumption fails for the pressure field.

The reason is rooted in a fundamental operator-level distinction. A fluid heated from below or beside responds in its temperature field and in its pressure field. These two responses are coupled through the equation of state and the momentum balance, yet they are governed by different mathematical operators with different Green's functions, different characteristic times, and different spatial footprints. Temperature is transported: it is advected, diffused, and filamented by the flow. Pressure, in the incompressible limit, is not transported at all; it is the Lagrange multiplier of the divergence-free constraint, determined instantaneously by a global elliptic solve. At finite compressibility, that instantaneous adjustment is replaced by acoustic waves propagating at speed c, but the adjustment remains fast compared to the advective timescale whenever the Mach number is small.

This distinction is elementary at the level of the governing equations. Its consequences for stochastic closure schemes have not been stated explicitly. A stochastic parameterization that only matches the energy spectrum of the unresolved pressure correction necessarily discards the phase coherence and boundary-awareness enforced by the elliptic operator—a global, non-local constraint that randomization destroys.

This paper shows that the operator split imposes a hard constraint on stochastic closures that spectrum-matching cannot satisfy. We introduce a spectrum-matched surrogate test that isolates the geometric requirement of the elliptic operator. We apply it to a Boussinesq projection method, and show that a phase-randomized field with the identical power spectrum—the structural move of spectrum-matched stochastic closures—fails to repair the divergence constraint by four orders of magnitude. Figure 1 shows the construction in one glance—the true correction, its spectrum, the surrogate, and the divergence ledger—so that the paper's central result is visible before the supporting measurements. The construction of a closure that respects this constraint is the subject of the companion paper (Saifelden 2026).

We establish the operator split across five independent settings before delivering the surrogate test: field data, direct numerical simulation, a controlled linear-acoustics crossover demonstration, a nonlinear Mach-number sweep, and reanalysis. The measurements show that the split is large enough to matter in real flows. The surrogate test then proves that any closure that discards elliptic geometry is structurally incomplete.

2 Phenomenological evidence: the two-clock fingerprint in nature

Before testing whether closures can capture the operator split, we establish that the split is large enough to matter in real and simulated flows. The following subsections present evidence from field measurements, direct numerical simulation, controlled solvers, and reanalysis. The observed signature is consistent across every setting: pressure and temperature respond to the same forcing through different operators with different timescales and different spatial footprints.

2.1 Temporal decoupling in field data

We first ask whether the two clocks are distinguishable in time-series measurements at fixed locations. If pressure and temperature respond to the same forcing on the same timescale, their spectra should be similar and their bulk correlation should be tight. If they respond through different operators, their spectra should differ qualitatively and their correlation should be weak and regime-dependent.

2.1.1 Single-site measurement: NEON eddy covariance

We use the NEON bundled eddy-covariance product (DP4.00200.001) at the Wind River Experimental Forest site (WREF; 45.82°N, 121.95°W; 53 m canopy height) for January 2020 at 30-minute resolution: 1488 intervals, of which 1097 carry valid pressure and 1180 valid temperature (NEON 2020).

Figure 2 shows diurnal composites of incoming shortwave radiation, air temperature, sensible heat flux, and barometric pressure. Three features are apparent.

First, the source–response chain is causal and ordered: incoming shortwave leads air temperature by approximately two hours and the sensible heat flux by approximately three hours. The temperature signal is driven locally by solar heating.

Second, the spectral characters of the two fields are qualitatively different. Temperature variance is concentrated at the 24-hour diurnal line. Pressure variance is dominated by multi-day synoptic variability with a distinct secondary peak at the 12-hour semidiurnal atmospheric tide, visible as twin daily maxima in the raw composite.

Third, the bulk daily correlation between pressure and temperature is weak: $r(P, T) \approx +0.07$ ($R^2 \approx 0.005$). This is far from the tight positive coupling that a constant-density ideal-gas relation would predict. The mismatch is accommodated by density fluctuations of order one percent; the gas law holds locally, but bulk pressure does not track bulk temperature.

A seasonal cross-check using July 2020 at the same site reproduces the qualitative fingerprint at the opposite solar extreme. The temperature diurnal signal becomes even cleaner (semidiurnal-to-diurnal power ratio drops from 7.3% in winter to 1.0% in summer), pressure retains its semidiurnal tide component (dominant at 11.9 h, 8.5% of its diurnal power), and the bulk daily correlation shifts to $r = -0.49$—the summer thermal-low regime, in which hot afternoons lower surface pressure. The dominant variability timescale in pressure is approximately 168 h (multi-day synoptic), with an observed slope of -0.024 kPa/K compared to the constant-density lock slope of +0.334 kPa/K. The sign and magnitude of the bulk correlation are set by the synoptic regime; the spectral separation between the two fields is not.

2.1.2 Multi-station measurement: the latitude fingerprint of a global mode

A single tower cannot distinguish "pressure is noisy" from "pressure responds to the whole atmosphere." To make this distinction, we examine one-minute barometric and thermometric data from eight ASOS stations spanning 24.6°–47.4°N (Iowa Environmental Mesonet archive; Q1 2020). Confidence intervals are computed via daily moving-block bootstrap ($n = 500$).

Table 1 summarizes the key measurements. The 24-hour temperature amplitude varies with local insolation and surface type: it is largest at the interior desert site (PHX, 5.06°C) and smallest at the marine sites (SEA, 1.79°C; EYW, 1.51°C). It does not order monotonically with latitude.

The 12-hour pressure-tide amplitude, by contrast, decreases monotonically from 1.10 hPa at 24.6°N

Table 1: Semidiurnal pressure-tide amplitude, diurnal temperature amplitude, and daily P–T correlation at eight ASOS stations, Q1 2020. Bootstrap 95% confidence intervals ($n = 500$) are given for the pressure-tide amplitude and daily correlation. The ratio to the Haurwitz migrating-tide climatology $A_H = 1.16 \cos^3 \varphi$ is shown in the final column.

Station	φ (°N)	$S_{1,T}$ (°C)	$S_{2,P}$ [95% CI] (hPa)	Daily P–T [95% CI]	$S_{2,P}/A_H$
EYW	24.6	1.51	1.103 [1.013, 1.191]	-0.16 [-0.30, -0.03]	1.26
MIA	25.8	2.73	1.077 [0.979, 1.169]	-0.05 [-0.18, +0.08]	1.27
MSY	30.0	2.75	1.018 [0.878, 1.169]	-0.40 [-0.54, -0.24]	1.35
DFW	32.9	3.28	0.957 [0.794, 1.141]	-0.48 [-0.59, -0.34]	1.39
PHX	33.4	5.06	0.924 [0.861, 1.008]	-0.20 [-0.29, -0.12]	1.37
DSM	41.5	2.73	0.628 [0.398, 0.965]	-0.56 [-0.67, -0.41]	1.29
CAR	46.9	3.03	0.439 [0.181, 0.879]	-0.42 [-0.52, -0.30]	1.19
SEA	47.4	1.79	0.337 [0.262, 0.529]	-0.06 [-0.21, +0.09]	0.94

to 0.34 hPa at 47.4°N. This is the classical signature of the global solar semidiurnal tide—a planetary normal-mode response whose amplitude is largest where the forcing projects onto the low-latitude waveguide (Chapman & Lindzen 1970). The trend correlation with the migrating-tide climatology $A_H = 1.16 \cos^3 \varphi$ hPa (Haurwitz 1956) is 0.98. Station totals fall in the range 0.9–1.4$\times A_H$, consistent with the expected contribution of non-migrating components (Dai & Wang 1999).

Spectral coherence between pressure and temperature at the solar harmonic lines (24 and 12 h) is high, ranging from 0.82 to 0.98 across stations. In off-line bands (9–11 h and 13–20 h), coherence drops to 0.09–0.30, above the phase-randomized-surrogate 95% level at all eight stations. The daily bulk pressure–temperature correlation is weak and negative (range -0.56 to -0.05), reflecting the dominance of the semidiurnal tide over diurnal thermal forcing.

The interpretation is direct: pressure at a fixed site carries the signature of a planetary-scale resonance, while temperature carries the signature of the local solar cycle. The two fields share a forcing (the sun) but respond through different operators with different spatial footprints. Station-level daily pressure–temperature correlations range from -0.05 to -0.56, with bootstrap 95% confidence intervals excluding any tight positive lock: the bulk coupling is set by the synoptic regime, not by a gas-law link between the two fields.

Band-resolved coherence sharpens this statement. The magnitude-squared coherence between hourly pressure and temperature is high (0.82–0.98) at the solar harmonic lines, where both fields are phase-locked to the same astronomical forcing, and above the phase-randomized surrogate 95% level at all eight stations. In the off-line sub-diurnal band (9–11 h and 13–20 h), coherence collapses to the surrogate noise floor. Away from the shared solar clock, nothing locks the two fields together.

2.2 Spatial decoupling in direct numerical simulation

Field sensors measure time series at a point; they cannot resolve spatial structure. A direct numerical simulation can. We use the two-dimensional Rayleigh–Bénard configuration from The Well dataset (Ohana et al. 2024): $Ra = 10^6$, $Pr = 1$, 512×128 grid, periodic in the horizontal, no-slip walls, with velocity, buoyancy, and pressure co-resolved on the same grid.

Table 2: Spatial scale comparison between buoyancy and pressure in Rayleigh–Bénard DNS at $Ra = 10^6$, $Pr = 1$. Absolute values and ratios are shown for integral length and Taylor microscale. The high-wavenumber spectral power ratio at $k \approx 10$ is approximately 10^4.

Metric	Buoyancy	Pressure	Ratio
Integral length L	0.195	0.207	1.06
Taylor microscale λ	0.253	0.607	2.40
High-k power at $k \approx 10$	1.0	10^{-4}	10^4

Table 3: Taylor-microscale ratio (λ_p/λ_θ) and integral-scale ratio (L_p/L_θ) as functions of Ra, from The Well $Pr = 1$ Rayleigh–Bénard dataset. Values are medians with 5–95% ranges over 30 snapshots per Ra (artifact `15b_rb_ra_sweep.json`).

Ra	λ_p/λ_θ	L_p/L_θ
10^6	2.58 [1.98–3.59]	1.02 [0.88–1.40]
10^7	2.94 [2.54–3.65]	1.25 [1.03–1.85]
10^8	4.45 [3.27–5.88]	1.75 [1.42–2.11]

The central question is whether pressure and buoyancy, driven by the same convective instability, develop the same spatial structure. The answer is no, and the difference is precisely what the operator analysis predicts.

At large scales, the two fields agree: both are organized by the convection rolls, and their integral length scales are nearly identical (0.195 for buoyancy, 0.207 for pressure; ratio 1.06). At small scales, they diverge sharply. The Taylor microscale of pressure is 2.40 times that of buoyancy (0.607 versus 0.253), and the high-wavenumber spectral power of buoyancy exceeds that of pressure by a factor of approximately 10^4 at $k \approx 10$.

Both numbers are direct consequences of the inverse Laplacian. The factor $1/k^2$ in Fourier space suppresses pressure power by $\sim 1/k^4$ relative to its source, so pressure remains smooth and domain-spanning while buoyancy is advected into thin filaments with intense small-scale gradients. Figure 4 shows a snapshot; Figure 5 shows the spectra.

The microscale ratio is not an artifact of a single Rayleigh number. Sweeping $Ra = 10^6$, 10^7, and 10^8 using The Well's $Pr = 1$ test files (5 trajectories $\times$ 6 statistically steady snapshots each) yields the results in Table 3. The Taylor-microscale ratio grows from 2.58 at $Ra = 10^6$ to 4.45 at $Ra = 10^8$, while the integral-scale ratio remains order unity. The high-wavenumber spectral gap widens by approximately 70$\times$ across the Ra sweep. As Ra increases and buoyancy plumes sharpen toward finer scales, the inverse Laplacian continues to hold pressure at the large scales. The operator split widens with turbulence intensity rather than closing.

To isolate the operator contrast from turbulence statistics, we apply a single localized source to both the elliptic and parabolic operators in a developed cavity flow. The elliptic response spreads over the entire domain: 79% of its magnitude lies beyond one source-width from the origin. The parabolic response at matched time remains a confined Gaussian blob. Boundary geometry edits the global pressure field instantly and only the local temperature field (Fig. 6).

The closure-relevant consequence of this operator split is directly measurable in the field data. Using the same NEON tower, we bin the transport-efficiency correlations for momentum and heat by

Table 4: Monin–Obukhov stability-bin transport-efficiency correlations from NEON WREF. Medians and 5–95% ranges are shown over usable periods (winter: $n = 602$; summer: $n = 1030$). Momentum-to-heat transport-efficiency ratio varies by stability, demonstrating differential response to shear-to-buoyancy ratio.

Stability class	ζ range	n (winter)	r_{uw}	$\lvert r_{wT} \rvert$	Ratio
Strongly unstable	< -1.0	95	0.182 [0.12–0.26]	0.209 [0.15–0.28]	0.87
Unstable	-1.0–-0.1	73	0.324 [0.28–0.37]	0.150 [0.11–0.19]	2.17
Near-neutral	-0.1–0.1	121	0.312 [0.27–0.35]	0.095 [0.07–0.12]	3.28
Stable	0.1–1.0	153	0.225 [0.18–0.27]	0.200 [0.17–0.23]	1.13
Strongly stable	> 1.0	160	0.156 [0.12–0.20]	0.203 [0.18–0.23]	0.77

Monin–Obukhov stability classes. Table 4 shows the momentum-to-heat transport-efficiency ratio swings by a factor of 4.3 in winter (0.77 to 3.28) and 6.0 in summer (0.43 to 2.60), demonstrating that momentum transport (shaped by the global pressure field) and heat transport (shaped by local buoyant plumes) respond differently to changes in the shear-to-buoyancy ratio. A single fixed-Pr_t diffusivity is structurally unable to reproduce this variation. Just as one eddy diffusivity cannot capture this stability-dependent transport split, one spectrum cannot capture the phase-dependent elliptic split: the surrogate test of Section 4 shows that matching power spectra is not enough to reproduce constraint-preserving action.

3 Dynamical mechanism: why the fast clock is not just "fast noise"

The phenomenological evidence establishes that the operator split is large enough to matter. Before delivering the surrogate test, we demonstrate the mechanism in controlled settings where every parameter is known. We define the slow clock by the advective timescale $\tau_{\text{slow}} \sim L/U$ and the fast clock by the acoustic adjustment timescale $\tau_{\text{fast}} \sim L/c$; in the incompressible limit $c \to \infty$ the fast clock becomes instantaneous and the hyperbolic wave equation degenerates into the elliptic constraint. These demonstrations serve a specific purpose: they show that the fast clock converges to a deterministic global field with a unique geometry. Randomizing its phases is therefore guaranteed to fail.

3.1 The hyperbolic-to-elliptic crossover: deterministic relaxation to a global field

A minimal two-dimensional linear-acoustics solver makes the relationship between the hyperbolic and elliptic operators concrete. An initial pressure perturbation radiates as a ring at speed c and decays by geometric spreading. Under sustained forcing, the compressible pressure relaxes to the same field as the elliptic Poisson solve, with a final relative error of $\sim 2 \times 10^{-3}$ for every sound speed tested.

The relaxation time scales as

$$t_{90} \approx 2.3\,\frac{L}{c}, \tag{1}$$

Table 5: Mach-number sweep in the nonlinear compressible solver. $\langle KE_{\text{dil}}\rangle/\langle KE_{\text{sol}}\rangle$ is the time-averaged ratio of dilatational to solenoidal kinetic energy. The pressure residual is the RMS difference between the acoustic-averaged compressible pressure and the elliptic Poisson pressure, normalized by the RMS pressure.

M	$KE_{\text{dil}}/KE_{\text{sol}}$	Pressure residual
0.05	1.5×10^{-4}	8.1×10^{-4}
0.10	5.9×10^{-4}	3.2×10^{-3}
0.20	2.3×10^{-3}	1.4×10^{-2}
0.40	9.3×10^{-3}	1.0×10^{-1}

and the error histories for all sound speeds collapse onto a single curve when time is rescaled as ct/L (Fig. 7). The sound speed sets the clock; the final state is independent of it. As $c \to \infty$, the adjustment becomes instantaneous and the hyperbolic equation degenerates into the elliptic constraint.

This establishes that the fast clock is not stochastic noise: it is a deterministic relaxation to a unique global field determined by the elliptic operator. A stochastic parameterization that randomizes the fast correction discards this deterministic structure.

3.2 The Mach-number sweep: energy scaling and the elliptic limit

A fully nonlinear two-dimensional isothermal compressible Navier–Stokes solver (pseudo-spectral, periodic, with $p = c^2\rho$ so that c is an explicit parameter) is initialized from a Taylor–Green vortex at $Re \approx 600$. The velocity field is Helmholtz-decomposed into solenoidal (vortical, slow) and dilatational (acoustic, fast) components at each time step.

Table 5 summarizes the Mach-number sweep. The dilatational energy fraction scales as $\sim M^2$, vanishing in the incompressible limit. The time-averaged pressure (averaged over the fast acoustic period) converges to the elliptic Poisson pressure with correlation approaching unity and residual shrinking as M^2.

Two clocks coexist in one flow: the vortical energy decays on the slow advective timescale while the acoustic energy oscillates on the fast period (Fig. 8). Averaging over the fast clock acts as a low-pass filter that recovers the elliptic limit—an operation we will repeat on the real atmosphere in Section 5.

This justifies why stochastic closures try to model the fast clock as a small random forcing—its energy is $O(M^2)$. But as we show next, energy level alone is not enough.

4 The geometric failure of spectrum-matched stochastic closures

The preceding sections have established that the fast pressure clock is a deterministic, global, boundary-aware field with a specific elliptic geometry. We now ask: If a stochastic parameterization only knows the energy spectrum of the fast correction, can it reconstruct the elliptic geometry?

The answer is no.

4.1 The standard projection method as a two-clock synthesis

The standard projection (fractional-step) method advances a Boussinesq flow by a local drift followed by a global elliptic correction (Chorin 1968; Leray 1934):

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t \left[-(\mathbf{u}\cdot\nabla)\mathbf{u} + \nu\nabla^2\mathbf{u} + (b - \langle b \rangle)\hat{\mathbf{y}} \right], \tag{2}$$

$$\mathbf{u}^{n+1} = \mathbb{P}\,\mathbf{u}^*, \tag{3}$$

$$b^{n+1} = b^n + \Delta t \left[-(\mathbf{u}\cdot\nabla)b + \kappa\nabla^2 b \right], \tag{4}$$

where $\mathbb{P} = I - \nabla(\nabla^2)^{-1}\nabla\cdot$ is the Leray projector. The drift step (Eq. 2) is a local, Markovian update—the parabolic clock. The projection step (Eq. 3) is a single global Poisson solve—the elliptic clock acting as a constraint.

Three results follow.

First, the intermediate velocity $\mathbf{u}^*$ violates mass conservation at RMS $\nabla\cdot\mathbf{u}^* \approx 2.7\times10^{-2}$. A single projection reduces this to $\approx 2\times10^{-15}$—machine zero (Fig. 10).

Second, replacing the projection correction by a phase-randomized field with the identical power spectrum (Parseval-exact)—the structural move of spectrum-matched stochastic closures (Hasselmann 1976; Berner et al. 2017)—yields a pointwise correlation with the true correction of median $|r| = 0.07$ (95th percentile 0.20, over an ensemble of $n = 100$ surrogates). The RMS divergence after the surrogate correction is 3.7×10^{-2} [3.5–3.8 $\times10^{-2}$, 95% interval]—no better than applying no correction at all (2.7×10^{-2}).

Third, the physical interpretation is that the fast clock's action on the slow dynamics is a global elliptic correction with a definite geometric structure, not a stochastic variance to be re-injected with the correct energy. Matching a power spectrum does not reconstruct a boundary-aware, globally coupled elliptic operator.

4.2 The target: stochastic parameterization schemes

The surrogate test establishes a general principle: spectral equivalence is not dynamical equivalence. Two fields can share identical power spectra while producing radically different outcomes under a constrained operator. The test targets the specific hypothesis that matching the power spectrum of an unresolved correction is sufficient to reproduce its constraint-preserving action. We do not claim that all stochastic parameterizations fail; rather, we show that spectrum-matching alone is insufficient for elliptic constraints: the phase coherence enforced by the elliptic operator ∇^2 is destroyed by randomization, and no spectral surrogate restores it.

The failure is not subtle. A field that shares the exact power spectrum with the true projection correction but has scrambled phases fails to satisfy the divergence constraint by four orders of magnitude. The phase structure is not a second-order correction; it is the essential geometric requirement of the elliptic operator. The surrogate test is performed in a two-dimensional periodic geometry, where the Poisson solve is a global inversion over the entire box; the structural conclusion—that phase coherence enforced by the elliptic operator is destroyed by randomization—holds for any domain in which incompressibility is imposed by a global Poisson solve, including bounded geometries with nontrivial boundary conditions (Section 6.3).

Table 6: Ratio of divergent to rotational kinetic energy (daily mean) from NCEP/NCAR Reanalysis 1 (R1) and NCEP–DOE Reanalysis 2 (R2). The minimum at 500 hPa corresponds to the classical level of non-divergence.

Level	R1: $KE_{\text{div}}/KE_{\text{rot}}$	R2 cross-check
850 hPa	4.0%	4.5%
500 hPa	0.7%	0.8%
250 hPa	1.0%	1.1%

5 The two clocks in the real atmospheric wind field

The preceding sections establish the operator split in controlled and semi-controlled settings. We now ask whether the same structure is visible in the real atmosphere.

5.1 Method

Horizontal winds from the NCEP/NCAR Reanalysis 1 (2.5° resolution; Kalnay et al. 1996) are Helmholtz-decomposed on the sphere via spherical-harmonic inversions $\nabla^2\psi = \zeta$ and $\nabla^2\chi = \delta$ into a rotational (non-divergent, balanced) component and a divergent (fast) component. The same decomposition is applied to the NCEP–DOE Reanalysis 2 (Kanamitsu et al. 2002) as an independent cross-check.

Horizontal divergence is not a model artifact: by three-dimensional continuity, it is the genuine footprint of vertical motion. The diagnostic is therefore the energy split, not the existence of divergence.

5.2 Results

Table 6 gives the ratio of divergent to rotational kinetic energy at three pressure levels, from both reanalyses.

Four features emerge.

First, the rotational wind dominates the kinetic energy at every level and every resolved scale. Splitting by spherical-harmonic degree l, the divergent branch sits one to two decades below the rotational branch for all $l \leq 35$ (Fig. 13).

Second, the vertical structure is physically meaningful: the divergent fraction is smallest near 500 hPa—the classical level of non-divergence (Holton 2004)—and larger in the boundary layer and near the jet outflow (Fig. 14).

Third, the cross-check with Reanalysis 2 reproduces the same ordering and the same 500-hPa minimum, with relative differences of 15% or less. The ratio is a property of the analyzed atmosphere, not of one assimilation system.

Fourth, time-averaging acts as a low-pass filter on the fast clock: the divergent fraction is smaller in the daily mean than in the 6-hourly data at every level (at 500 hPa, 0.7% versus 1.1%). Averaging scrubs the fast divergent clock—exactly as acoustic averaging recovered the elliptic pressure in Section 3.2—consistent with the mechanism of Section 3.

6 Discussion

6.1 Summary of measurements

Five independent measurements—field observation, direct numerical simulation, a controlled linear-acoustics crossover demonstration, a nonlinear Mach-number sweep, and reanalysis—consistently reveal the same structure: pressure and temperature respond to the same forcing through different operators with different timescales and different spatial footprints. The temporal signatures differ (local diurnal versus planetary semidiurnal); the spatial signatures differ (filamentary versus smooth, with a $1/k^4$ spectral gap); the dynamical demonstrations show the mechanism (hyperbolic-to-elliptic crossover, M^2 energy scaling, projection as synthesis); and the real atmosphere exhibits the expected energy partition with the correct vertical structure.

The surrogate test then proves that spectrum-matching alone cannot reconstruct the elliptic geometry. This is a hard constraint that any stochastic closure must respect.

6.2 What a valid closure must include

The surrogate test identifies three non-negotiable requirements for any closure that aims to represent the fast pressure clock:

Global coupling (non-local Green's function). The elliptic operator ∇^2 is non-local in physical space. A closure that treats the fast correction as a local stochastic process cannot capture the global coupling imposed by the incompressibility constraint.

Boundary-aware phase structure. The phase coherence of the elliptic correction is determined by the domain boundaries and the source distribution. Spectrum-matching discards this phase structure; a valid closure must preserve it.

Zero divergence to machine precision. The divergence constraint is not a statistical requirement; it is an exact kinematic constraint. A closure that does not enforce $\nabla \cdot \mathbf{u} = 0$ to machine precision is structurally incomplete.

The construction of a closure that respects these constraints is the subject of the companion paper (Saifelden 2026).

6.3 Scope and limitations

All numerical demonstrations are two-dimensional and periodic; in particular, the surrogate test of Section 4 is performed in a periodic geometry, and the structural conclusion—that phase coherence enforced by the elliptic operator is destroyed by randomization—is argued to hold for any domain where incompressibility is imposed via a global Poisson solve, including bounded geometries with nontrivial boundary conditions. No three-dimensional regularity claim is made. The reanalysis diagnostics are limited to spherical-harmonic degree $l \leq 35$ and are based on model-assimilated products, not raw observations. The field measurements span two months at the seasonal extremes (NEON) and one season at eight sites (ASOS); they establish the fingerprint of the operator split but do not constitute a global climatology.

Table 7: Falsification criteria for the main claims. Each claim can be rejected by a single counterexample meeting the stated condition.

Claim	Kill condition
E1: Temporal split	Pressure and temperature show identical diurnal power ratios
E2: Spatial split	Taylor microscale ratio approaches unity at high Ra
E3: Transport efficiency	Momentum-to-heat ratio is constant across stability classes
E4: Crossover scaling	t_{90} does not scale as L/c in compressible solver
E5: Spectrum matching	Phase-randomized surrogate achieves machine-zero divergence
E6: Reanalysis split	Divergent kinetic energy exceeds 10% at any level

6.4 Falsification criteria

The main claims of this paper can be falsified as follows:

All claims remain open to stronger tests with three-dimensional simulations, higher-resolution reanalysis, or extended field campaigns.

6.5 Reproducibility

All figures and numerical results regenerate from the public repository at `https://github.com/abdh0saifelden2-debug/K`. Each analysis is a single command with fixed seeds, runnable on CPU. The repository's test suite (656 passing tests at draft time) covers the analysis modules. Input datasets are public: NEON DP4.00200.001 (data.neonscience.org), ASOS one-minute data via the Iowa Environmental Mesonet (mesonet.agron.iastate.edu), The Well Rayleigh–Bénard dataset (polymathic-ai), and NCEP/NCAR Reanalysis 1 via NOAA PSL OPeNDAP (psl.noaa.gov). See Appendix A for the full command table.

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A Reproducibility

Table 8 lists the commands that regenerate each result in this paper.

Table 8: Reproducibility commands. All are runnable from the repository root with fixed seeds and CPU-only execution.

Result	Command
§2.1.1 NEON diurnal composites	`python general_two_clocks/run_analysis.py`
§2.1.1 Seasonal cross-check	`python general_two_clocks/run_neon_compare.py`
§2.1.2 ASOS latitude sweep	`python general_two_clocks/asos/fetch_iem.py && python general_two_clocks/run_asos_ci.py`
§2.2 RB DNS split	`python general_two_clocks/run_rb.py`
§2.2 *Ra* sweep	`python general_two_clocks/run_rb_ra_sweep.py`
§2.2 Green's function	`python general_two_clocks/run_elliptic_demo.py`
§2 Stability bins (E3)	`python general_two_clocks/run_stability.py`
§3.1 Linear crossover	`python general_two_clocks/run_compressible.py`
§3.2 Nonlinear Mach sweep	`python general_two_clocks/run_compressible_ns.py`
§4 Projection / surrogate	`python general_two_clocks/run_boussinesq.py`
§5 Reanalysis split	`python general_two_clocks/run_reanalysis.py`
R2 cross-check	`python general_two_clocks/run_reanalysis_r2.py`

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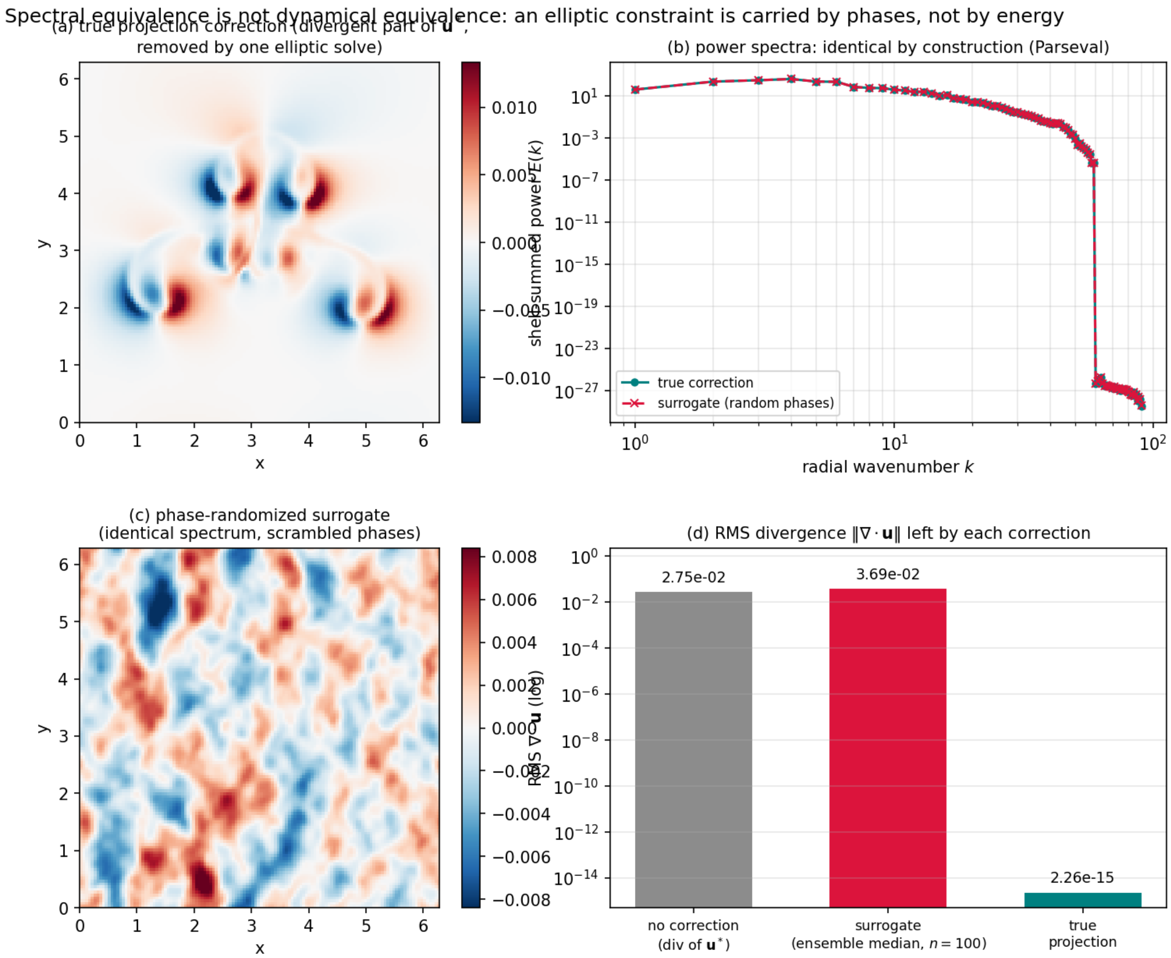

Figure 1: The surrogate test in one figure. (a) The true projection correction: the *divergent* part of the intermediate velocity $\mathbf{u}^*$ that a single elliptic solve removes. (b) Power spectra of the true correction and of a phase-randomized surrogate—identical by construction (Parseval-exact). (c) The surrogate correction field: the same spectrum, scrambled phases. (d) RMS divergence left by no correction (2.7×10^{-2}), by the surrogate (ensemble median 3.7×10^{-2}, $n = 100$), and by the true projection (2×10^{-15}). Spectral equivalence is not dynamical equivalence: a field that matches the correction's power spectrum does not reproduce its action as a constraint.

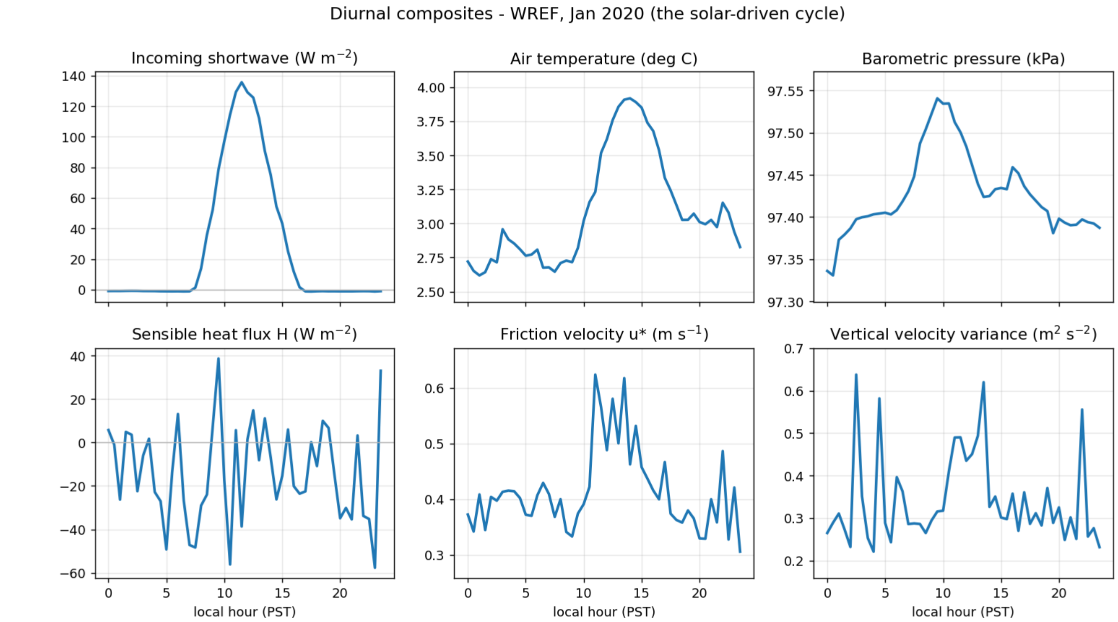

Figure 2: NEON WREF, January 2020: diurnal composites of incoming shortwave radiation, air temperature, sensible heat flux, and barometric pressure. Temperature exhibits a single clean 24-hour signal driven by local solar heating. Pressure shows a twin-peak semidiurnal tide superimposed on multi-day synoptic variability. Confidence intervals are computed via daily moving-block bootstrap ($n = 500$).

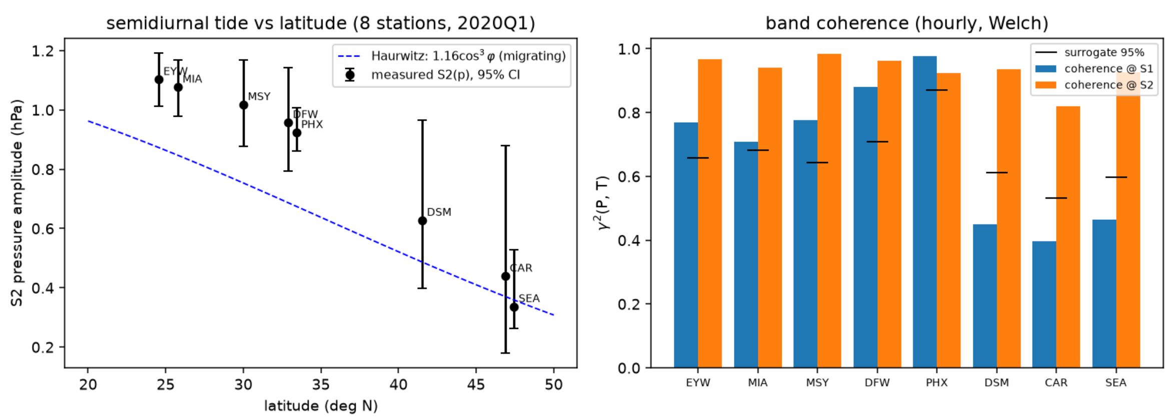

Figure 3: Left: semidiurnal pressure-tide amplitude versus latitude at eight ASOS stations with bootstrap 95% confidence intervals, compared to the Haurwitz migrating-tide climatology (dashed curve). The monotonic equator-to-pole decay is the signature of a global resonance. Right: band-resolved coherence between hourly pressure and temperature—high at the shared solar lines, at the surrogate noise floor off-line.

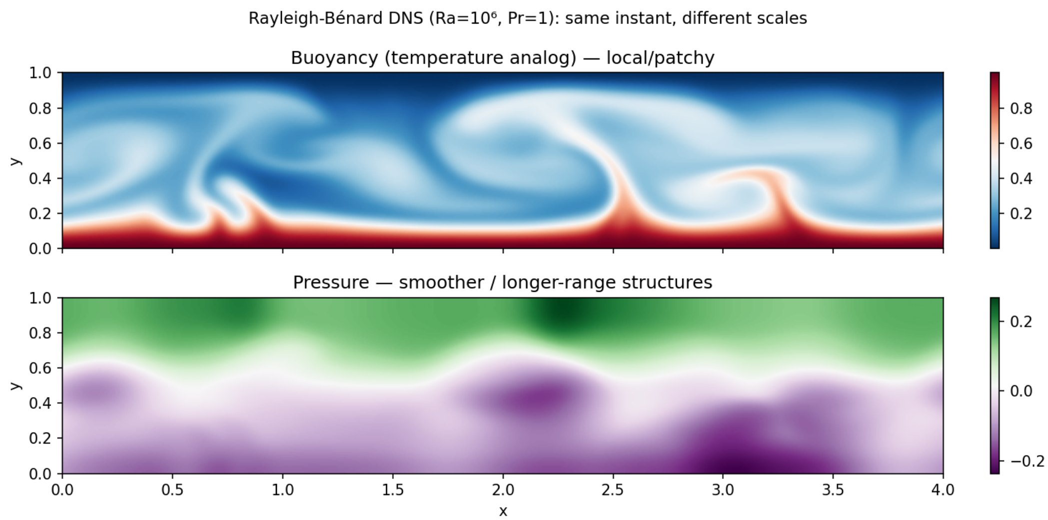

Figure 4: Rayleigh–Bénard DNS snapshot at $Ra = 10^6$: buoyancy (left) is filamentary and sharp, organized into thin plumes; pressure (right) is smooth and domain-spanning. Both fields are from the same simulation at the same instant.

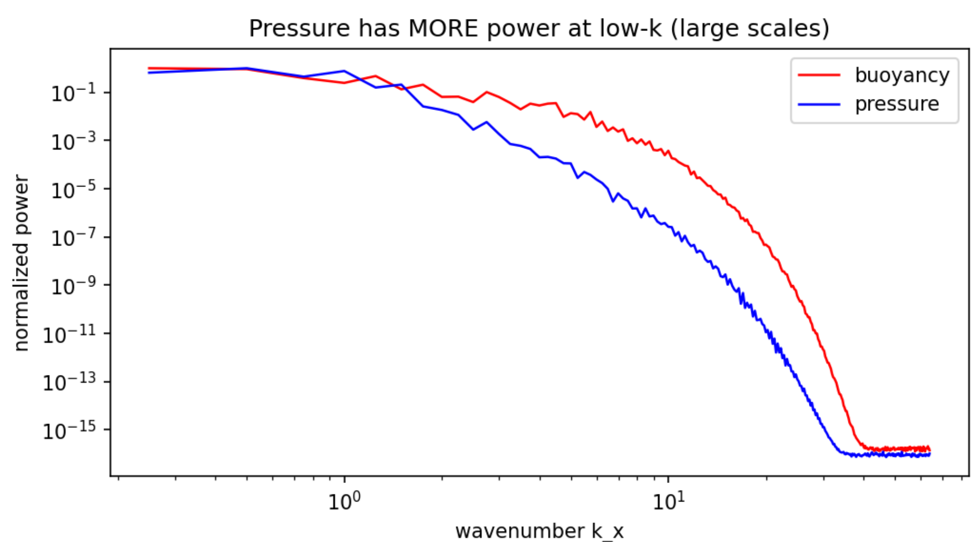

Figure 5: Spatial spectra of buoyancy and pressure from the Rayleigh–Bénard DNS. The energy-containing scales coincide; at high wavenumber, buoyancy exceeds pressure by $\sim 10^4$, consistent with the $1/k^4$ suppression of the inverse Laplacian.

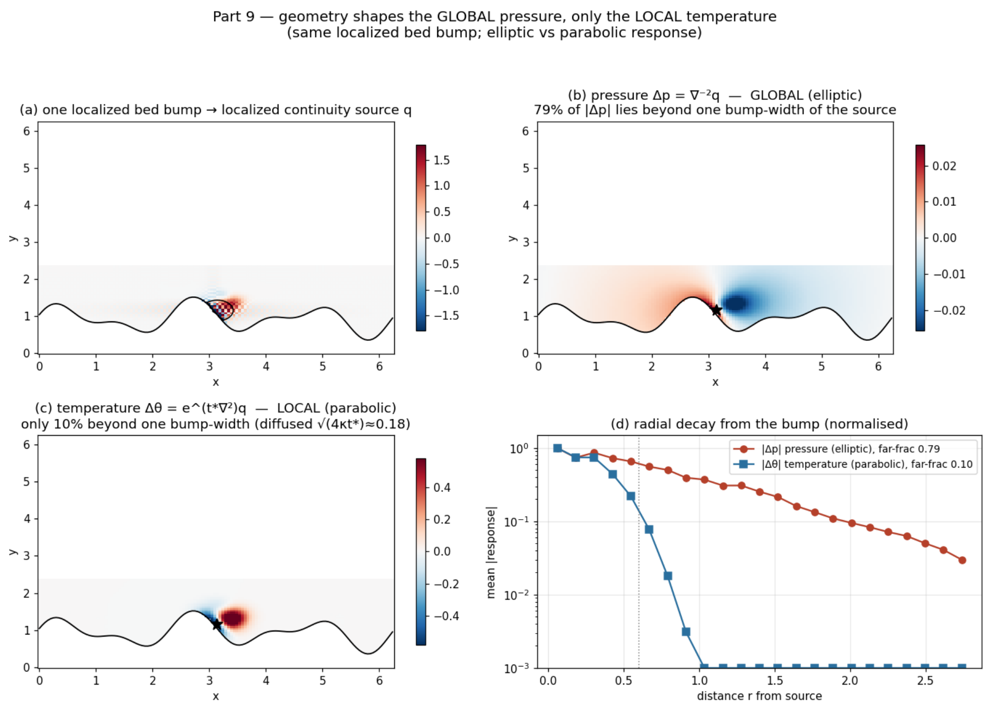

Figure 6: Response to an identical localized source applied to both operators. The elliptic (pressure) response spreads globally, with 79% of its magnitude beyond one source-width. The parabolic (heat-kernel) response remains confined.

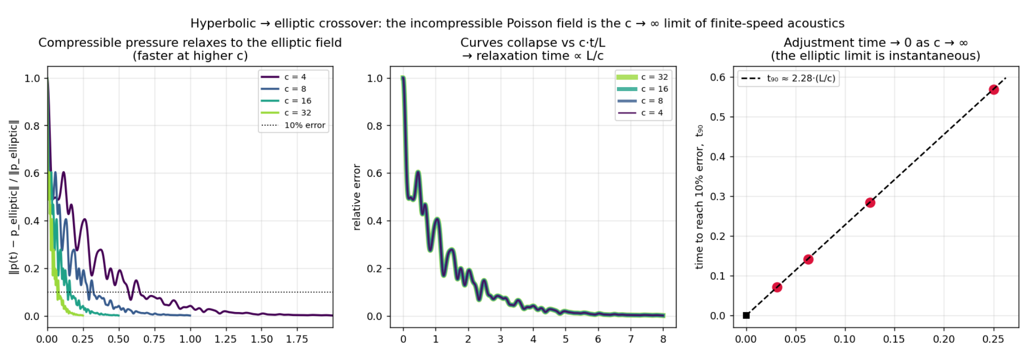

Figure 7: Relaxation of the compressible pressure field to the elliptic (Poisson) solution for several sound speeds. All curves collapse in the rescaled time ct/L, with $t_{90} \approx 2.3\, L/c$.

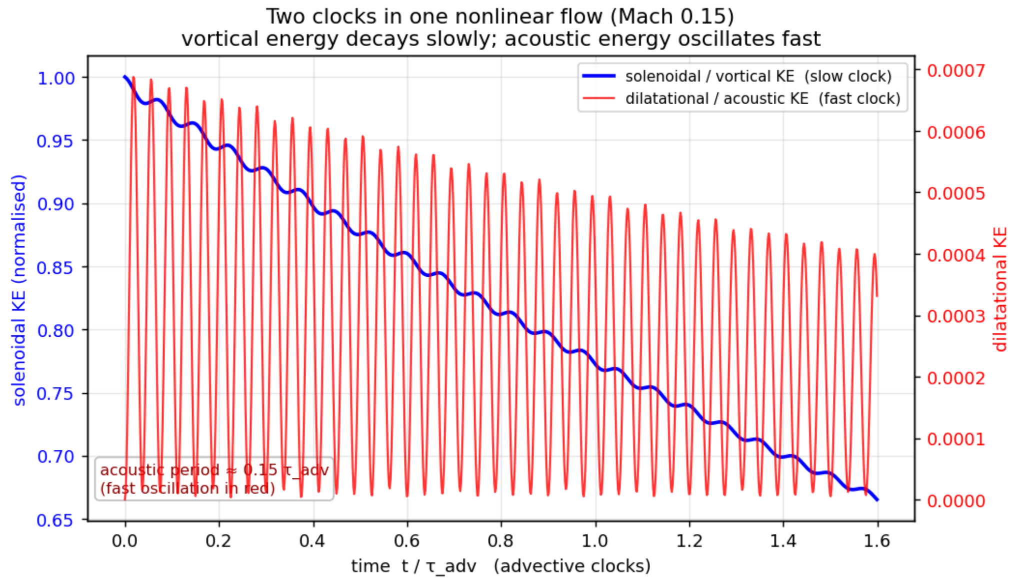

Figure 8: Nonlinear compressible box at $M = 0.1$: slow vortical energy decay with fast acoustic oscillation superimposed. Both clocks are present in one flow.

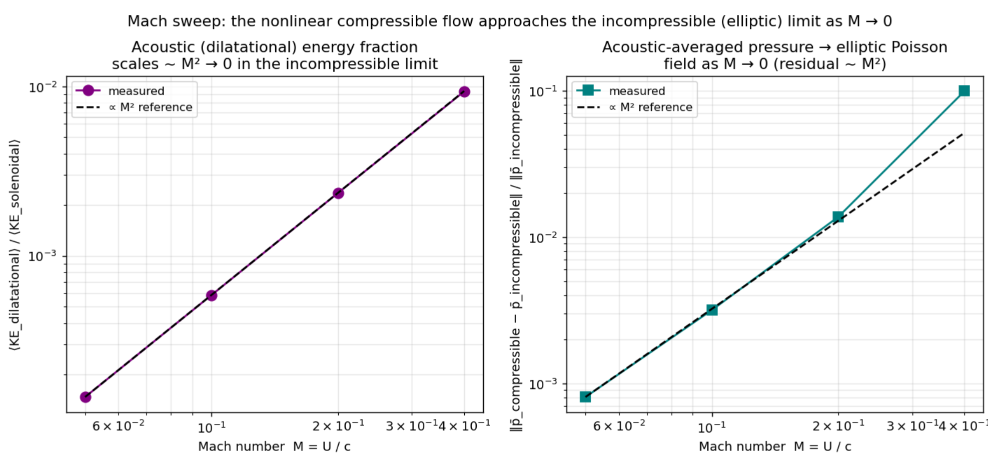

Figure 9: Mach-number sweep: dilatational energy fraction (circles) and acoustic-averaged pressure residual (squares) both scale as $\sim M^2$ (dashed line).

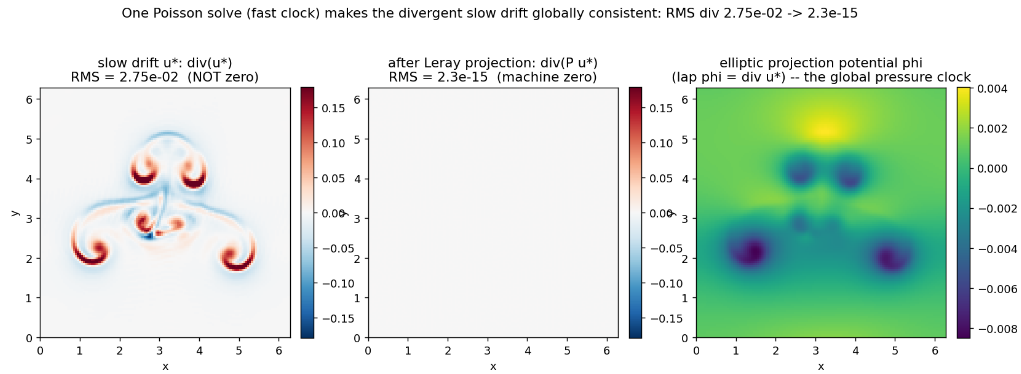

Figure 10: One projection time step, dissected. Top: the divergent intermediate velocity $\mathbf{u}^*$ (RMS $\nabla \cdot \mathbf{u}^* = 2.7 \times 10^{-2}$). Middle: the global elliptic potential ϕ satisfying $\nabla^2\phi = \nabla \cdot \mathbf{u}^*$. Bottom: the projected, solenoidal velocity $\mathbf{u}^{n+1}$ (RMS $\nabla \cdot \mathbf{u}^{n+1} = 2 \times 10^{-15}$).

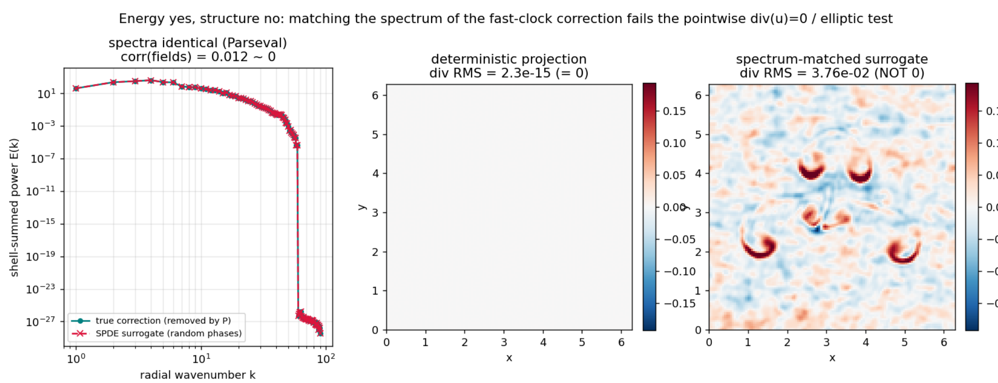

Figure 11: Spectrum-matched surrogate test. A phase-randomized field with the identical power spectrum as the true projection correction (Parseval-exact) fails to reproduce the correction pointwise: ensemble-median correlation $|r| = 0.07$ ($n = 100$), and the divergence constraint is left unrepaired (RMS $\nabla \cdot \mathbf{u} = 3.7 \times 10^{-2}$, compared to 2×10^{-15} for the true projection).

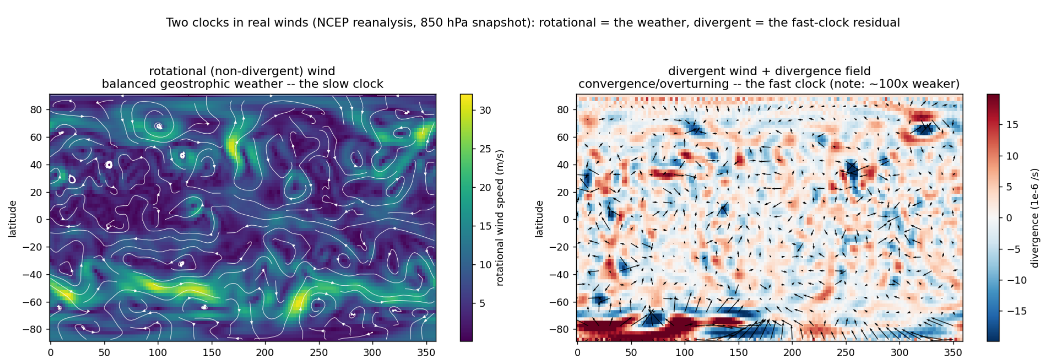

Figure 12: NCEP/NCAR winds, one analysis day. Left: rotational wind (synoptic highs, lows, and jets). Right: divergent wind (weak, confined to convergence and overturning zones).

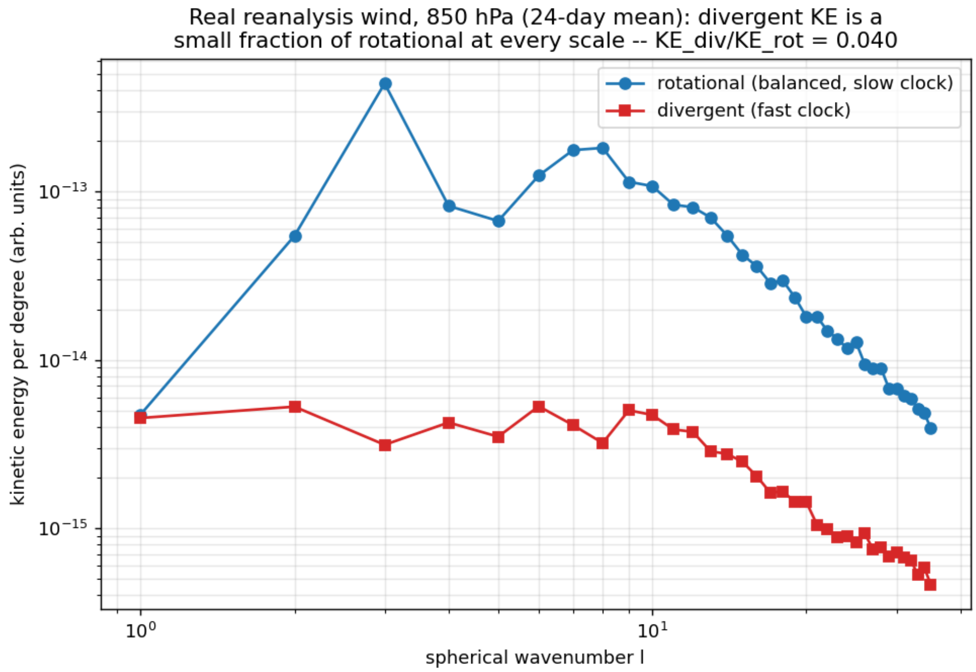

Figure 13: Kinetic-energy spectra by spherical-harmonic degree l: the divergent branch sits one to two decades below the rotational branch at every resolved scale.

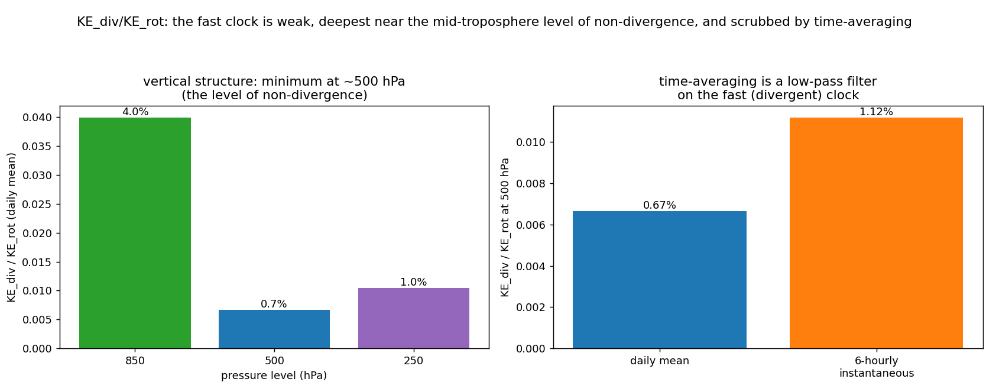

Figure 14: Vertical structure of $KE_{\mathrm{div}}/KE_{\mathrm{rot}}$: minimum at the level of non-divergence (∼500 hPa). Six-hourly values (open symbols) exceed daily-mean values (filled symbols), demonstrating that time-averaging acts as a low-pass filter on the fast clock.